

Dispensing Practical II

PCT 212 • 200 Level Pharm.D • Second Semester

A Practice Workbook

An original revision aid by the EverythingABUAD student team | everythingabuad.com

How to use this workbook. These are **original self-test items** written from scratch by the EverythingABUAD student team, modelled on the PCT 212 syllabus (mixtures, linctuses, mouthwashes & gargles, paints, nasal & ear drops, inhalations, and the supporting calculations and theory). They are **not past papers** and are **not a prediction of any test**. Work each question on your own first — do the calculation, build the table of quantities, and reason out the ‘why’ — then check the worked Answer Key at the back. Always cross-check quantities, doses and definitions against your lecturer’s current course outline and the current pharmacopoeia.

Section 1 — Mixtures (Diffusible & Indiffusible Solids)

1. You are to dispense an antacid/adsorbent chalk mixture. Using the master formula below (per 100 mL), prepare enough to send **150 mL** to Master D. Tanko (allow a small excess and prepare 160 mL).

Chalk	5 g
Tragacanth powder	2% w/v
Chloroform water, double-strength	10% v/v
Water	to 100 mL

Send 150 mL. Label: 10 mL to be taken three times daily.

- Draw up the table of quantities for the 160 mL you will prepare.
- Is chalk a diffusible or an indiffusible solid, and what is the role of the tragacanth?
- State the container and the auxiliary label required.

2. A compound magnesium antacid mixture is prescribed. The per-dose (10 mL) formula is given below. Send **five doses**; prepare six doses to allow for measuring loss.

Magnesium trisilicate	500 mg
Light magnesium carbonate	500 mg
Sodium bicarbonate	5% w/v
Peppermint water	to 10 mL (one dose)

Label: one 10 mL spoonful three times daily.

- Prepare the table of quantities for six doses.
- Are these solids diffusible or indiffusible? Does the mixture need a suspending agent? Give a reason.

3. In a **table**, distinguish between **diffusible** and **indiffusible** solids as used in oral mixtures. Compare their behaviour on standing, their need for a suspending agent, and give **three examples of each**.

Section 2 — Linctuses

4. Send **6 doses of Codeine Linctus BP**; the dose is one 5 mL spoonful twice a day. Prepare about 40 mL to allow for loss.

- (a) What total volume does 6 doses represent, and why do you prepare a little extra?
- (b) State three properties/actions desired in a linctus.
- (c) Why is a linctus sipped and swallowed slowly, and not diluted with water?
- (d) Why is it unwise to administer codeine linctus frequently or for a prolonged period?

5. A diabetic patient, Mr. M. Zaha, needs a cough linctus. Why is a **Diabetic Codeine Linctus** preferred for him over the ordinary Codeine Linctus? What is changed in its formula, and why does that change not spoil the linctus action?

6. In a table, compare a **linctus** and a **mixture** as oral liquid dosage forms (vehicle, viscosity, purpose, how it is taken, usual dose volume).

Section 3 — Mouthwashes & Gargles

7. A phenol mouthwash is prescribed. Send **50 mL** to Lady M. Bintu; label 10 mL, suitably diluted, as a mouthwash.

Liquefied phenol	3% v/v
Potassium hydroxide solution	3% v/v
Amaranth solution	1% v/v
Water	to 50 mL

- (a) Draw the table of quantities for 50 mL.
- (b) State the role of each ingredient.
- (c) Why are oily or mucilaginous substances not used in gargles and mouthwashes?
- (d) What cautions apply because the preparation contains phenol?

8. Prepare **100 mL** of a potassium permanganate solution such that when **10 mL is diluted to 200 mL** it yields a **1 in 5000** solution for use as a gargle. You are to use a **0.5% w/v stock solution** of potassium permanganate.

- (a) What concentration must the 100 mL you dispense be?
- (b) How much stock solution and how much water do you use?

9. A potassium chlorate and phenol gargle is ordered. The master formula (per 100 mL) is below. Send **160 doses** of 5 mL to Male Ward A; label 5 mL when required.

Potassium chlorate	3% w/v
Patent blue V	0.001 g
Liquefied phenol	1.5% v/v
Water	to 100 mL

- (a) What total volume is needed and what is the table of quantities?
- (b) State the role of each ingredient.

Section 4 — Paints

10. A Brilliant Green and Crystal Violet skin paint is prescribed (master per 100 mL below). Send **80 mL** to Lady M. Bintu; prepare 100 mL.

Brilliant green	0.5% w/v
Crystal violet	0.5% w/v
Alcohol (90%)	50% v/v
Water	to 100 mL

- (a) Give the table of quantities for 100 mL.
- (b) Comment on the types of paint, their differences and typical examples.
- (c) State the container and labelling.

11. (a) Distinguish between **Magenta Paint (Castellani's Paint)** and **Compound Iodine Paint (Mandl's Paint)**. (b) Why is such a paint labelled "Store in a cool place"? (c) Distinguish between an **official** and an **unofficial** preparation.

Section 5 — Nasal & Ear Drops

12. Ephedrine nasal drops are prescribed (master per 100 mL below). Send **20 mL** to Adaku Igwe; prepare 25 mL. Dose: one drop into each nostril.

Ephedrine hydrochloride	0.5% w/v
Chlorbutol	0.5% w/v
Sodium chloride	0.5% w/v
Water	to 100 mL

- (a) Give the table of quantities for 25 mL, and the role of each ingredient.
- (b) Why are oily liquids not used as vehicles for nasal drops?
- (c) Why is this preparation not recommended for repeated/prolonged use?

13. Sodium bicarbonate ear drops are prescribed (master per 100 mL below). Send **30 mL** to Lady A. Sule; prepare 40 mL. Dose: three drops into the affected ear three times a day.

Sodium bicarbonate	5% w/v
Glycerol	30% v/v
Purified water	to 100 mL

- (a) Give the table of quantities for 40 mL.
- (b) What is the role of glycerol (and of sodium bicarbonate) in this ear drop?
- (c) Why should ear drops be warmed to body temperature before use?

Section 6 — Inhalations

14. Menthol and Eucalyptus Inhalation is prescribed (master per 100 mL below). Prepare and dispense an appropriate quantity.

Menthol	2% w/v
Eucalyptus oil	10% v/v
Light magnesium carbonate	7.5% w/v
Water	to 100 mL

- (a) What is the role of the light magnesium carbonate in this preparation?
- (b) State the container and the directions/label a patient should be given.

Section 7 — Pharmaceutical Calculations & Theory

15. Chloroform water. (a) You have only **double-strength** chloroform water. How much of it, and how much water, will you use to produce **75 mL of single-strength** chloroform water? (b) Concentrated Chloroform Water BP is diluted **one part to forty** to give ordinary (single-strength) chloroform water. How much water must you add to **50 mL** of the concentrate to make single-strength chloroform water?

16. Aromatic waters. (a) Using double-strength rose water only, how much will you use to make **60 mL** of single-strength rose water? (b) A concentrated aromatic water is diluted **one part to forty** for single strength. How much water must you add to **50 mL** of concentrated dill water to make **double-strength** dill water?

17. Strength conversions. (a) Express 0.5% w/v as a '1 in X' ratio. (b) Express '1 in 5000' as a percentage strength. (c) Paracetamol paediatric elixir contains 120 mg in 5 mL — express this as % w/v.

18. List and describe the **types of water** used in pharmaceutical preparation, giving a use for each.

19. A Calamine Lotion is prescribed (formula below). (a) What are the classes of lotion? (b) State the role of each ingredient and identify the suspending agent. (c) How is the product classified, and how is it labelled?

Calamine	150 g
Zinc oxide	50 g
Bentonite	30 g
Sodium citrate	5 g
Liquefied phenol	5 mL
Glycerol	50 mL
Water	to 1000 mL

20. With a suitable example for each, explain the following terms: **analgesic, antipyretic, astringent, antiseptic, carminative, demulcent, rubefacient, sedative.**

Answer Key

Worked answers, grouped to match the questions above.

Section 1 — Mixtures (Diffusible & Indiffusible Solids)

1. Scale factor = $160 \div 100 = 1.6$. Multiply every quantity by 1.6:

Chalk	$5 \text{ g} \times 1.6 = 8 \text{ g}$
Tragacanth powder	$2\% \text{ of } 160 \text{ mL} = 3.2 \text{ g}$
Chloroform water, double-strength	$10\% \text{ of } 160 \text{ mL} = 16 \text{ mL}$
Water	to 160 mL

(b) Prepared (heavy) chalk is an **indiffusible** solid — it is dense, settles quickly and cakes at the bottom, so an even dose cannot be poured without help. The **tragacanth (2% w/v) is the suspending agent**: it raises the viscosity of the vehicle so the chalk stays dispersed long enough for an accurate dose to be measured.

Chloroform water is the vehicle/preservative and flavour.

(c) **Container**: plain amber (coloured fluted not required) medicine bottle. **Auxiliary label**: “Shake well before use.”

2. Six doses = $6 \times 10 \text{ mL} = 60 \text{ mL}$ total.

Magnesium trisilicate	$500 \text{ mg} \times 6 = 3 \text{ g}$
Light magnesium carbonate	$500 \text{ mg} \times 6 = 3 \text{ g}$
Sodium bicarbonate	$5\% \text{ of } 60 \text{ mL} = 3 \text{ g}$
Peppermint water	to 60 mL

(b) Magnesium trisilicate and light magnesium carbonate are **diffusible** solids (light, fine powders that disperse through the vehicle and settle only slowly, re-dispersing on shaking). **No suspending agent is needed** — a shake before use gives a uniform dose. Sodium bicarbonate is soluble. Still label “Shake well.”

3. Diffusible vs indiffusible solids:

Feature	Diffusible	Indiffusible
Nature	Light, fine, low-density insoluble powders	Dense/heavy insoluble powders
On standing	Settle slowly; re-disperse easily on shaking	Settle fast and cake; hard to re-disperse
Suspending agent	Not required	Required (e.g. tragacanth, compound tragacanth powder, acacia)
Examples	Light magnesium carbonate, light kaolin, magnesium trisilicate	Aspirin, phenobarbitone, prepared chalk, calamine, zinc oxide

Section 2 — Linctuses

4. (a) $6 \times 5 \text{ mL} = 30 \text{ mL}$ is the dispensed volume; you prepare ~40 mL because a viscous syrup clings to the conical measure and mortar, so an excess ensures the full 30 mL can be delivered.

(b) Desired properties: **demulcent** (soothes the inflamed pharyngeal mucosa); **cough-suppressant** (codeine is a central antitussive); and a **sweet, viscous, palatable syrup vehicle** that both improves acceptance and prolongs mucosal contact.

(c) It is sipped slowly and swallowed **undiluted** so the thick syrup coats the throat and gives a prolonged local demulcent/soothing effect; diluting or drinking water afterwards washes the film off and loses that action.

(d) Codeine is an opioid — frequent use risks **constipation, respiratory depression, tolerance and dependence**, and it suppresses a productive cough (retaining secretions). It is therefore used sparingly and avoided in young children.

5. The ordinary linctus vehicle is **syrup (sucrose / invert sugar)** — a large sugar load unsuitable for a diabetic. The diabetic version replaces the sugar syrup with a **non-sugar viscous, sweet vehicle** (e.g. sorbitol solution / glycerol with a thickener such as tragacanth). Because the sweetness and the **viscosity** are retained, the linctus still coats the throat and acts as a demulcent, but without raising blood glucose or contributing to dental caries.

6. Linctus vs mixture:

Feature	Linctus	Mixture
Vehicle	Syrup / sweet viscous base	Aqueous (chloroform / peppermint water)
Viscosity	Thick, viscous	Thin, mobile
Purpose	Soothe throat / relieve cough (demulcent)	Deliver a dose of drug(s) — systemic/local
How taken	Sipped slowly, swallowed undiluted	Swallowed, may be diluted; shake if a suspension
Dose volume	Small (~5 mL)	Larger (10–15 mL)

Section 3 — Mouthwashes & Gargles

7. (a) Table of quantities for 50 mL:

Liquefied phenol	3% of 50 mL = 1.5 mL
Potassium hydroxide solution	3% of 50 mL = 1.5 mL
Amaranth solution	1% of 50 mL = 0.5 mL
Water	to 50 mL

(b) **Liquefied phenol** = antiseptic and mild local analgesic (the active); **potassium hydroxide solution** = alkali to give a suitable (alkaline) pH and aid solution; **amaranth solution** = red colourant (identifies the product and warns it is not to be swallowed); **water** = vehicle.

(c) Oily/mucilaginous substances **coat the mucosa with a clinging film**, feel unpleasant, do not rinse away cleanly and **impair the antiseptic's contact and spread** over the throat; they are also hard to expel. Gargles/mouthwashes therefore use thin aqueous vehicles.

(d) Phenol is **corrosive and a poison**: draw the examiner's attention to it and sign the poison book, handle with care (avoid skin/eye contact, wear gloves), dispense in a coloured bottle, and label "Not to be swallowed — for

rinsing the mouth only.”

8. (a) ‘1 in 5000’ = 1 g in 5000 mL = 0.0002 g/mL = **0.02% w/v** at the point of use. Diluting 10 mL up to 200 mL is a **20-fold** dilution, so the dispensed solution must be $0.02\% \times 20 = \mathbf{0.4\% \text{ w/v}}$.

(b) 100 mL of 0.4% w/v contains 0.4 g potassium permanganate. From a 0.5% stock (0.5 g/100 mL): volume of stock = $(0.4 \div 0.5) \times 100 = \mathbf{80 \text{ mL of stock}}$, made up to 100 mL with **20 mL of water**. Check: 10 mL of 0.4% = 0.04 g; in 200 mL that is 0.04 g/200 mL = 1 g/5000 mL = 1 in 5000. ✓

9. (a) 160 doses $\times$ 5 mL = **800 mL**. Scale factor from the 100 mL master = 8:

Potassium chlorate	3% of 800 mL = 24 g
Patent blue V	0.001 g $\times$ 8 = 0.008 g
Liquefied phenol	1.5% of 800 mL = 12 mL
Water	to 800 mL

(b) **Potassium chlorate** = astringent/antiseptic for mouth and throat (active); **liquefied phenol** = antiseptic and local analgesic (a poison — handle with care); **patent blue V** = colourant/identifier; **water** = vehicle. Coloured bottle; “Not to be swallowed — gargle only.”

Section 4 — Paints

10. (a) Table of quantities for 100 mL:

Brilliant green	0.5% of 100 mL = 0.5 g
Crystal violet	0.5% of 100 mL = 0.5 g
Alcohol (90%)	50% of 100 mL = 50 mL
Water	to 100 mL

(b) **Paints** are liquids applied to the skin or mucous membranes. **Skin paints** use a volatile solvent (alcohol/acetone) that evaporates to leave the active as a film — e.g. this Brilliant Green & Crystal Violet paint, Magenta Paint. **Throat / mucous-membrane paints** are viscous and **glycerol-based** (glycerol clings to and stays in contact with moist mucosa) — e.g. Compound Iodine Paint (Mandl’s), Phenol Glycerin. Difference: skin paints are volatile and quick-drying; throat paints are non-volatile, viscous and long-clinging.

(c) **Container:** coloured/amber bottle. **Labels:** “For external application only,” “Flammable” (contains alcohol), and a warning that the dyes **stain skin and clothing**.

11. (a) **Magenta Paint** is a **skin** paint (magenta/basic fuchsin with phenol, boric acid, resorcinol in acetone–alcohol) used as an antifungal/antibacterial for the skin. **Mandl’s Paint** is a **throat** paint (iodine + potassium iodide in a glycerol/alcohol base) applied to the throat and mouth. They differ in **base/solvent** (volatile acetone–alcohol vs viscous glycerol), **site of use** and **active ingredients**.

(b) “Store in a cool place” because the ingredients are **volatile / heat-sensitive** — acetone and alcohol evaporate readily (and are flammable) and iodine sublimates; cool storage limits loss of solvent/active and slows degradation.

(c) An **official** preparation has a formula and standard laid down in an official book (BP / BPC / pharmacopoeia). An **unofficial** preparation has no pharmacopoeial monograph — it is made to a prescriber’s own (extemporaneous) formula.

Section 5 — Nasal & Ear Drops

12. (a) Table for 25 mL (scale factor 0.25):

Ephedrine hydrochloride	0.5% of 25 mL = 0.125 g
Chlorbutol	0.5% of 25 mL = 0.125 g
Sodium chloride	0.5% of 25 mL = 0.125 g
Water	to 25 mL

Roles: **ephedrine hydrochloride** = sympathomimetic **decongestant** (active); **chlorbutol** = antimicrobial **preservative**; **sodium chloride** = adjusts **tonicity** toward that of nasal fluid to reduce irritation; **water** = vehicle.

(b) Oily vehicles (liquid paraffin, fixed oils) **impair the sweeping action of the nasal cilia**, are not miscible with watery nasal secretions, and — if aspirated into the lungs — can cause **lipoid (aspiration) pneumonia**. An aqueous vehicle is used.

(c) Repeated ephedrine use causes **rebound congestion (rhinitis medicamentosa)** and **tachyphylaxis** (diminishing effect), plus possible systemic sympathomimetic effects; so use is short-term only.

13. (a) Table for 40 mL (scale factor 0.4):

Sodium bicarbonate	5% of 40 mL = 2 g
Glycerol	30% of 40 mL = 12 mL
Purified water	to 40 mL

(b) **Glycerol** is a viscous, hygroscopic vehicle: it **holds the drops in the ear canal**, prolongs contact, and helps soften and loosen wax by retaining moisture. **Sodium bicarbonate** is alkaline and **softens/emulsifies ear wax (cerumenolytic)** so it can be removed.

(c) Cold drops entering the ear stimulate the labyrinth and can cause **dizziness / vertigo**; warming to body temperature avoids this and is more comfortable.

Section 6 — Inhalations

14. (a) The **light magnesium carbonate** is a **dispersing agent / distributor**. It adsorbs the volatile oils (menthol and eucalyptus oil) and lets them be **evenly distributed** through the water; when a spoonful is added to hot water the oils are then released and volatilise **steadily and uniformly** instead of floating on the surface as a separate oily layer.

(b) **Container**: amber bottle. **Directions/label**: “Add one teaspoonful to a pint of hot (not boiling) water and inhale the vapour”; “Shake well”; and “Not to be taken (swallowed).”

Section 7 — Pharmaceutical Calculations & Theory

15. (a) Single-strength = double-strength diluted with an equal volume of water. For 75 mL: use **37.5 mL double-strength** chloroform water and add **water to 75 mL** (i.e. 37.5 mL water). (b) Single-strength volume = $50\text{ mL} \times 40 = 2000\text{ mL}$; water to add = $2000 - 50 = 1950\text{ mL}$.

16. (a) For 60 mL single-strength: **30 mL double-strength** rose water + water to 60 mL. (b) ‘One part to forty’ gives single strength; **double strength** is one part to twenty. From 50 mL concentrate: double-strength volume = $50 \times 20 = 1000\text{ mL}$; water to add = $1000 - 50 = 950\text{ mL}$.

17. (a) $0.5\% \text{ w/v} = 0.5\text{ g in } 100\text{ mL} = 1\text{ g in } 200\text{ mL} = \mathbf{1\text{ in } 200}$. (b) $1\text{ in } 5000 = 1\text{ g in } 5000\text{ mL} = 0.0002\text{ g/mL} = \mathbf{0.02\% \text{ w/v}}$. (c) $120\text{ mg in } 5\text{ mL} = 0.12\text{ g in } 5\text{ mL} = 2.4\text{ g in } 100\text{ mL} = \mathbf{2.4\% \text{ w/v}}$.

18. Types of water: **Potable (drinking) water** — wholesome but not used directly in official manufacture. **Purified Water BP** (by distillation, ion-exchange or reverse osmosis) — the general solvent for most internal and external preparations. **Freshly boiled and cooled purified water** — where dissolved gases/microbes must be minimised (e.g. some solutions, eye lotions). **Water for Injections BP** — sterile and pyrogen-free, for parenteral products. **Distilled water** — purified by distillation; laboratory and preparative use.

19. (a) Lotions are external liquids applied without friction. Classes: **suspension / ‘shake’ lotions** (insoluble powders suspended, leaving a cooling powder film — e.g. Calamine Lotion), **solution (clear) lotions**, and **emulsion lotions** (e.g. Oily Calamine Lotion). (b) **Calamine** = mild astringent/antipruritic (and colour); **zinc oxide** = astringent and protective; **bentonite** = **suspending agent**; **sodium citrate** = stabiliser/peptising agent that keeps the suspension smooth; **liquefied phenol** = antipruritic/antiseptic; **glycerol** = humectant that keeps the skin moist and helps the powder adhere; **water** = vehicle. (c) It is a **suspension (shake lotion)**; label “Shake well before use” and “For external use only.”

20. **Analgesic** — relieves pain (paracetamol). **Antipyretic** — reduces fever (paracetamol/aspirin). **Astringent** — precipitates protein, contracts tissue, checks discharge (zinc oxide, tannic acid). **Antiseptic** — inhibits micro-organisms on living tissue (phenol, chlorhexidine). **Carminative** — relieves flatulence/gripping (peppermint oil, ginger). **Demulcent** — soothes and protects irritated mucosa (syrup, tragacanth, glycerol). **Rubefacient** — reddens skin by increasing local blood flow (methyl salicylate, camphor). **Sedative** — calms and reduces excitement (phenobarbitone).

Originality & IP Disclaimer

© EverythingABUAD | everythingabuad.com. All questions, explanations, and answers in this document were written from scratch by the EverythingABUAD student team as an original study aid. This document is not an official examination paper, is not affiliated with or endorsed by any university or college, and does not predict any future test. Always cross-check with your lecturer's current course outline. Free to use — not for resale.